Habitable — Project Proposal

October 20, 2026

Course: CS 501 E1, Mobile Application Development, Boston University, Fall 2026

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Git repository: github.com/Priyanshusingh09876/Habitable

Overview

Habitable is an Android app that helps renters in Boston record problems in their apartment, starting with no heat, so they can get them fixed. A cheap Bluetooth thermometer logs the room temperature and the app compares it against the Massachusetts heat rules. The tenant adds dated photos, sends a dated notice to the landlord, and exports one PDF packet that helps get a housing inspector to come out.

We presented a first draft on October 6. This proposal is built around the comments from that session, which are listed in the next section.

Android topics this project covers

Topic	Where it shows up
Jetpack Compose UI, Material 3	All six screens
Navigation with type-safe routes	Bottom bar plus issue and setup flows
ViewModel + StateFlow	One UI state per screen, survives rotation
Local persistence	Room as the source of truth, DataStore for settings
Device hardware	Bluetooth LE thermometer, CameraX
Background work	Foreground service for BLE, WorkManager for sync and reminders
Third-party API	Open-Meteo hourly weather
Networking and backend	Retrofit to our Node/Express + PostgreSQL API with JWT auth

Topic	Where it shows up
File generation and sharing	On-device PDF with PdfDocument, share sheet, email intent

Changes based on the October 6 feedback

Each comment from the presentation is now a concrete change in the plan.

Comment on October 6	What we changed	Where in this proposal
"How are you going to do the chart? You should have the chart early on."	Charts are in the MVP and in the first sprint. We use Vico, a chart library made for Jetpack Compose, on the Home and Sensors screens and draw the same data into the PDF.	Technical approach, Milestones
"What kind of evidence would be useful in a claim against the landlord?" (for example, a photo of the thermostat, or outdoor temperature)	We read MassLegalHelp and the State Sanitary Code and mapped each type of evidence to a feature. We added thermostat photos and outdoor weather from Open-Meteo.	Evidence research
"What does the Escalate button do?"	Escalate is now defined as three steps that follow the law: send a notice, request an inspection, then remind the landlord of the repair deadlines.	User journey
"When you say pool, what do you mean?" (pooling issues across a building)	Pooling is a stretch goal. Neighbors can opt in to share only a count, like "Unit 3B also has no heat." No readings or photos are shared.	Features by priority

Comment on October 6	What we changed	Where in this proposal
"Integrating with email to keep threads of communication with the landlord might be useful."	A landlord communication log is our first feature after the MVP. Every notice, reply and call is saved on the issue timeline.	Features by priority
"It's a limited-use app; once things get solved you won't use it anymore."	Move-in and move-out photo checklists give tenants a reason to keep the app between problems.	Features by priority
Proposal structure	We moved from two repositories to one, as the assignment requires.	Header

Problem and users

Massachusetts law sets clear heat rules, but tenants usually can't prove how bad things were. Under 105 CMR 410.180, every habitable room must be at least 68°F from 7 AM to 11 PM and 64°F overnight, from September 15 to May 31, and no warmer than 78°F. The temperature is measured 5 feet above the floor and more than 5 feet from an exterior wall.

Massachusetts law establishes minimum heating standards and specific procedures for addressing housing-code violations. Certain remedies under M.G.L. c.111 §127L involve five-day and fourteen-day timelines when statutory conditions are met. Habitable will present applicable guidance based on the issue and documented inspection status. A typical tenant has a few texts and a memory, which makes it hard to show how cold it was, for how long, and when the landlord was told.

Primary users: renters in Boston and Massachusetts dealing with slow or unresolved repairs, especially students and families.

Secondary users: people who receive the PDF and don't need the app, such as landlords, legal aid staff and housing inspectors.

Assumptions to test: tenants will keep a sensor on the wall for a few weeks, and a weeks-long temperature log is more convincing than a few texts.

Evidence research

MassLegalHelp tells tenants to do four things: notify the landlord in writing, take dated photos, record how long the problem lasted, and get a Board of Health inspection. The inspector's measurement is the official one, so our goal is a packet that helps get an inspection to happen. This is product research, not legal advice, and the app says so on every letter and PDF.

What helps a claim	Why it matters	Feature in Habitable
Dated written notice to the landlord	Creates the paper trail; the §127L repair deadlines run from written notice	Notice from a template, plus a log of when and how it was sent
How long and how cold	Inspectors ask about duration and severity	Readings that can't be edited, charted against the 68/64°F limit
Photo of the thermostat and radiator	Shows the heating system itself, not just the room	CameraX photo, with a SHA-256 hash taken at capture
Sensor placed where the law measures	5 ft up and 5 ft from an exterior wall (105 CMR 410.180)	Placement photo when the sensor is set up
Outdoor temperature	A cold room means more when it was freezing outside	Open-Meteo hourly weather on the same chart
ISD / Board of Health inspection	The official measurement; a certified violation starts the §127L clock	Escalate opens an inspection checklist and logs the request

User journey

The MVP is one complete path from measuring a problem to exporting the packet. If only this path works in December, we still have a usable app.

1. Create an account and add the apartment (address, unit, landlord contact).
2. Pair the BLE thermometer, or turn on simulated-sensor mode, and take a placement photo.
3. See the temperature chart against the legal limit on the Home screen.
4. The app opens a heat issue automatically when readings fall below the limit.

5. Add photos of the thermostat and radiator.
6. Send a dated notice to the landlord.
7. Follow the issue timeline and use Escalate for the next step.
8. Export the PDF packet.

Key screens

- **Home:** current temperature, the chart, open issues, and a chip when the room is below the limit.
- **Issue:** a timeline of sensor events, photos, notices and inspection requests, with the Escalate button.
- **Packet:** choose what goes into the PDF (chart, weather, photos, notice, flags, hash list) and export it.

What Escalate does

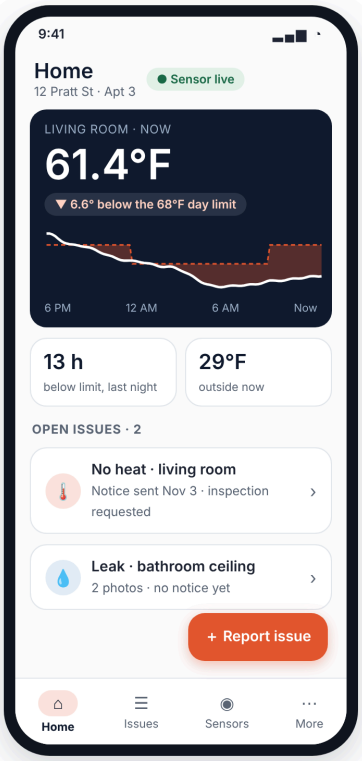
The button changes depending on where the issue is, and the order follows §127L:

1. **No notice sent yet:** the button says *Send notice*. It creates a dated letter from a template, sent by email or the share sheet and logged on the timeline.
2. **Notice sent, not fixed:** *Escalate* helps the tenant request an ISD / Board of Health inspection, with a checklist and a phone link, and logs the request.
3. **Violation certified:** the 5-day and 14-day repair deadlines start. *Escalate* sends the landlord a reminder, and WorkManager tracks the deadlines.

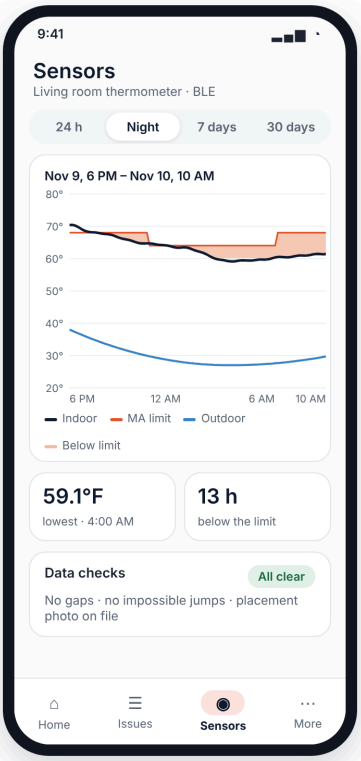
The app never files anything legal or with 311 for the tenant in the MVP.

UI design

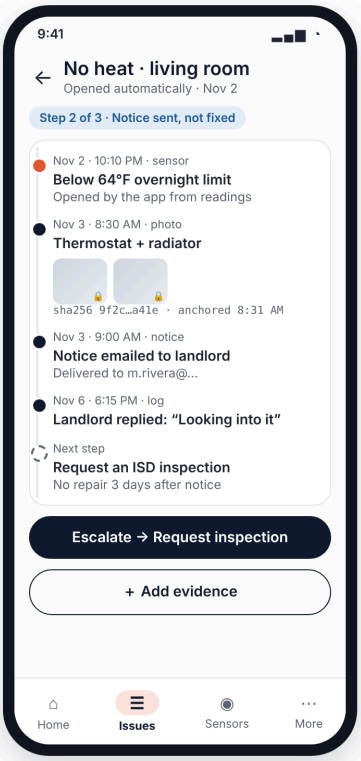
These are design mockups of the six MVP screens, not screenshots of a built app. They follow Material 3, use the same simulated night as our demo data, and show the chart on the first screen as requested on October 6.



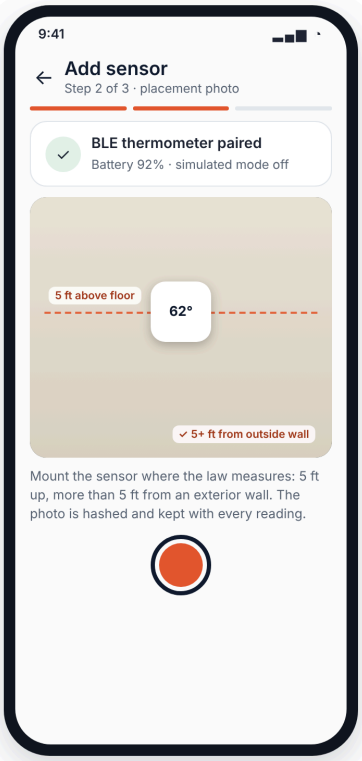
Home
Chart first, current reading, open issues



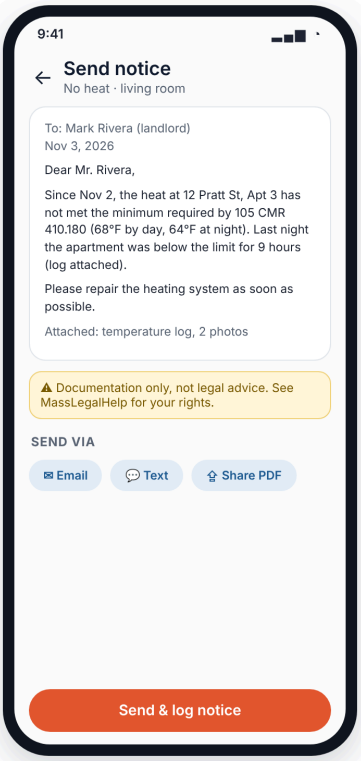
Sensors
Vico chart vs. the 68/64°F step limit



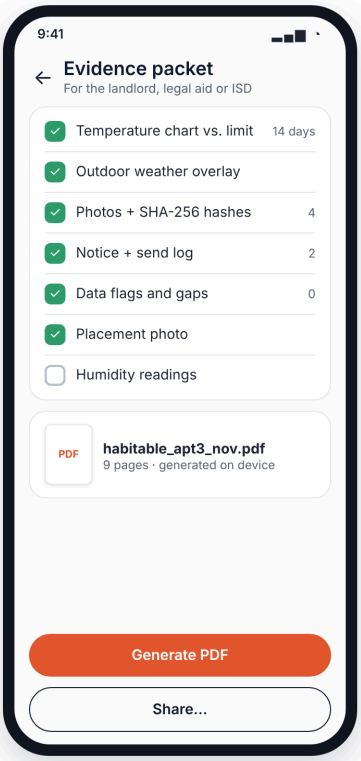
Issue timeline
Escalate always shows its next step



Add sensor
Placement photo at the legal spot



Send notice
Dated letter from a template



Evidence packet
One PDF an inspector can read

Screen	What it shows	Design decisions
Home	Current temperature, last night's chart, hours below the limit, outdoor temperature, open issues	The chart comes first; one tap reports a new issue
Sensors	Full chart with the 68/64°F step limit, outdoor line, violation shading, data checks	Same series is drawn into the PDF; data checks are always visible
Issue timeline	Every sensor event, photo, notice and reply in order, with photo hashes	The Escalate button names its next step, so it is never vague
Add sensor	Pairing status and a placement photo with 5 ft guides	Puts the sensor where the law measures before any reading counts
Send notice	Dated letter from a template, "not legal advice" banner, send options	Sending also logs the notice on the timeline
Evidence packet	Checklist of what goes into the PDF, the generated file	One PDF for the landlord, legal aid or ISD

Accessibility: text meets WCAG AA contrast, touch targets are at least 48 dp, colour is never the only signal (chips and the chart legend also use words), and all screens support TalkBack labels and system font scaling.

Features by priority

We build in three tiers, and nothing in a later tier starts until the MVP works end to end.

MVP (must ship)

1. BLE temperature and humidity logging, plus a simulated-sensor mode for development and demos
2. Heat rules: 68/64°F by time of day, the 78°F maximum, and the September 15 to May 31 season
3. Charts on Home, Sensors and in the PDF
4. Outdoor temperature from Open-Meteo on the same chart
5. CameraX photos with a SHA-256 hash taken at capture
6. Issue tracker with a timeline

7. Notice from a template, inspection request, and §127L deadline reminders
8. On-device PDF packet

Second priority (after the MVP works)

9. Landlord communication log: notices, replies and calls saved on the issue timeline
10. Mold-risk alerts from humidity
11. Move-in and move-out photo checklists

Stretch (only if time allows)

12. Noise log in decibels only, with no audio recorded
13. Building-wide counts: neighbors opt in to share only that their unit has the same problem
14. Open311 and address lookup
15. Encrypted backup of the evidence folder

Not in scope: legal advice, iOS, a live landlord portal, audio recording, and automating rent withholding.

Initial backlog

The MVP is split into user stories with acceptance criteria and estimates (S = under a day, M = 1 to 3 days, L = 3 to 5 days). Stories US-1, US-8 and US-9 are sprint 1, together with the chart (US-3) as requested.

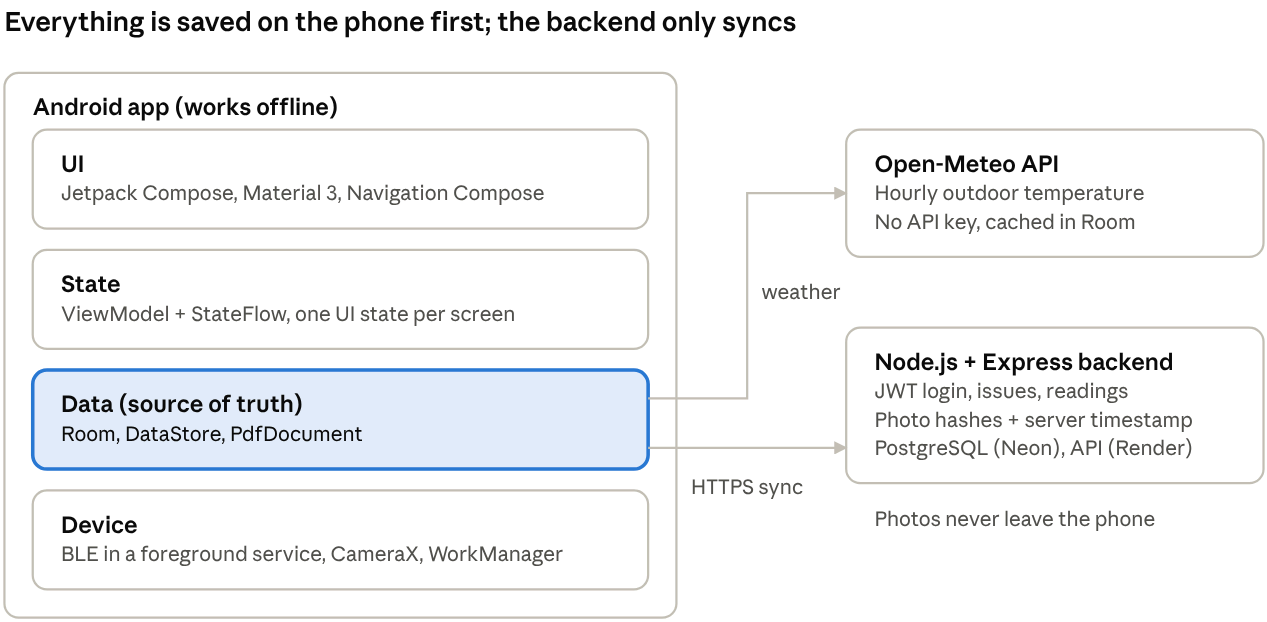
ID	User story	Acceptance criteria	Owner	Size
US-1	As a renter, I can create an account and add my apartment, so my evidence is tied to one address.	Sign up and log in with email and password (JWT); add address, unit and landlord contact; data survives app restart.	Both	S
US-2	As a renter, I can pair a BLE thermometer and take a placement photo, so readings are tied to a location.	Scan and pair one sensor; placement photo saved with a hash; readings stored in Room every 10 minutes.	Priyanshu	M

ID	User story	Acceptance criteria	Owner	Size
US-3	As a renter, I can see my temperature charted against the legal limit, so I know when it is too cold.	Vico chart shows indoor, outdoor and the 68/64°F step limit; shading only where below the limit; works offline.	Priyanshu	L
US-4	As a renter, my photos are time-stamped and hashed, so I can show they weren't edited later.	SHA-256 taken at capture; hash sent to `/evidence/anchor` when online; photos cannot be edited in the app.	Ankita	M
US-5	As a renter, I can open an issue and attach evidence.	Issue has category, description, status; photos and readings attach to it; a heat issue opens automatically after 2 hours below the limit.	Ankita	M
US-6	As a renter, I can send a dated notice and get reminders for next steps.	Notice built from a template with the "not legal advice" line; sent via email or share sheet and logged; WorkManager reminders at 5 and 14 days after a certified violation.	Ankita	M
US-7	As a renter, I can export one PDF for the landlord, legal aid or an inspector.	PDF includes the chart, weather, photos with hashes, notice log and data flags; generated on the phone; shareable.	Ankita	L
US-8	As developers, we can demo without hardware.	Simulated-sensor mode produces readings through the same pipeline as BLE; can replay a cold night on demand.	Priyanshu	S
US-9	As developers, we have the backend deployed.	Node/Express API with `/auth`, `/readings` and `/evidence/anchor` running on Render with Neon PostgreSQL.	Priyanshu	M

ID	User story	Acceptance criteria	Owner	Size
US-10	As a renter, my outdoor weather is shown next to indoor readings.	Open-Meteo hourly temperature for the apartment's location; cached in Room; loading, error and stale states handled.	Priyanshu	S

Technical approach

The Android app works offline and keeps everything on the phone first; the backend is small and only stores accounts, readings and photo hashes. We plan to use free-tier hosting services during development to minimize costs, with a low-cost BLE sensor as the only anticipated hardware expense.



app architecture · 4 layers on the phone, 2 services

Room on the phone is the source of truth; the app only reaches out for weather and to sync readings and photo hashes.

Area	Choice	Notes
Language and UI	Kotlin, Jetpack Compose, Material 3	Single activity; Home, Issues, Sensors and More in a bottom bar

Area	Choice	Notes
Navigation	Navigation Compose with type-safe routes	
State	ViewModel + StateFlow	One UI state per screen, survives rotation
Local data	Room (source of truth), DataStore for settings	Readings and photos are insert-only; there is no edit screen
Sensor	Bluetooth LE thermometer in a foreground service	Android 12+ limits background scanning; gaps are flagged
Camera	CameraX	SHA-256 hash taken at capture
Background work	WorkManager	Syncing readings and repair-deadline reminders
Charts	Vico	Same data series drawn into the PDF
Third-party API	Open-Meteo	Hourly outdoor temperature, no API key, cached in Room so charts work offline
PDF	Android PdfDocument	Generated on the phone
Networking	Retrofit + OkHttp, kotlinx.serialization	Readings uploaded in batches when online
Backend	Node.js + Express, JWT auth, PostgreSQL	Hosted on Render and Neon free tiers; laptop fallback with adb reverse for demos

Main endpoints: `/auth`, `/issues`, `/readings` (batch upload), `/notices`, and `/evidence/anchor`, which stores a server timestamp next to a photo's hash. The photos themselves never leave the phone.

Still researching: reading our chosen sensor's BLE data format, how often to sample without draining the battery, and keeping the foreground service alive on different phone brands.

Data model

Room stores the application's local operational data, including apartments, sensors, readings, issues, evidence metadata, notices, and timeline events. PostgreSQL stores server-side account information and synchronized records required for authentication, backup, and evidence anchoring. The two databases do not need identical schemas.

Table	Key fields	Notes
users	id, email, password_hash, name, created_at	Server only; password hashed with bcrypt
apartments	id, user_id, address, unit, city, lease_start, landlord_name, landlord_email	One user can have one or more apartments
sensors	id, apartment_id, ble_address, model, room, placement_photo_id, added_at	`ble_address` = "SIM" in simulated mode
temperature_readings	id, sensor_id, temp_f, humidity_pct, recorded_at, flagged	Insert-only; `flagged` marks jumps and gaps
heat_violations	id, sensor_id, started_at, ended_at, min_temp_f, required_temp_f, issue_id	Computed from readings and the 68/64°F rule
issues	id, apartment_id, category, title, description, status, opened_at, closed_at	Status: open, notice sent, inspection requested, certified, resolved
evidence_items	id, issue_id, kind, sha256, captured_at, server_timestamp, local_path	`local_path` stays on the phone; only the hash syncs
notices	id, issue_id, sent_at, method, repairs_start_by, repairs_finish_by	Deadlines filled once a violation is certified
timeline_events	id, issue_id, type, occurred_at, note	Drives the issue timeline and the communication log

Testing plan

We test the heat rules hardest, because a wrong rule would put a false number in a tenant's PDF.

- **Unit tests (JUnit):** the heat-rule engine at the edges (10:59 PM vs 11:01 PM, 6:59 AM vs 7:00 AM, September 14 vs 15, May 31 vs June 1), jump and gap detection, deadline math for 5 and 14 days.
- **Repository and Room tests:** readings and evidence are insert-only; offline writes sync once when the network returns.
- **UI tests (Compose testing):** the main flow from adding an apartment to exporting a PDF, run on the simulated sensor.
- **Backend tests (Jest + Supertest):** auth, batch upload of readings, and `/evidence/anchor` returning a server timestamp.
- **Manual device tests:** BLE sampling with the screen off for 8 hours on at least two phone brands, and a PDF check on a real printer.
- **CI:** GitHub Actions runs `assembleDebug`, unit tests and `npm test` on every pull request.

Data trust and privacy

We can't physically stop someone from putting the sensor in a fridge, so the app makes tampering visible instead of trying to prevent it.

- A placement photo is taken when the sensor is set up.
- Readings and photos can be added but never edited or deleted in the app.
- Impossible jumps (for example 70°F to 40°F within minutes) and gaps in the data are flagged.
- Flagged periods are printed in the PDF, not hidden.
- Outdoor weather gives a cross-check for the indoor readings.

SHA-256 hashes allow later changes to captured files to be detected, while server timestamps provide an additional record of when the hash was received. Neither mechanism independently verifies the authenticity of the photographed scene or the accuracy of sensor placement.

The inspector's measurement stays the official one. On privacy, photos and addresses stay on the phone, only hashes and metadata go to the server over HTTPS, and the app has no analytics SDKs. Building-wide sharing, if we build it, is opt-in and shares only a

count.

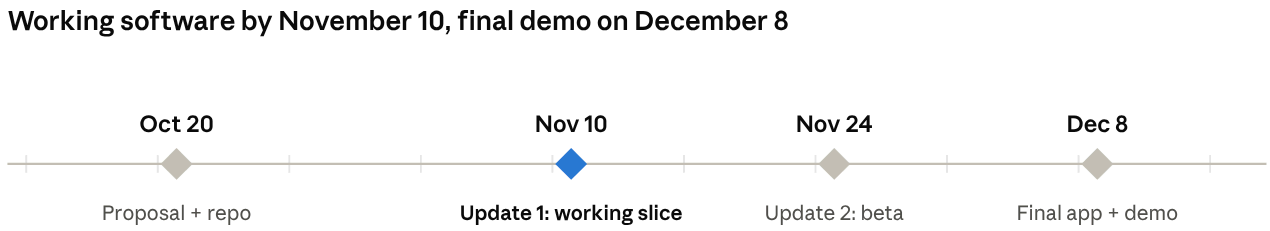
Team, process and milestones

Each of us owns one half of the app, and we share the parts that connect them.

Person	Owns
Priyanshu Singh	Sensing: BLE and the simulator, Room readings, the Massachusetts heat rules, Vico charts, Open-Meteo, WorkManager, backend schema and JWT auth
Ankita Patra	Evidence: issue tracker and timeline, CameraX with hashing, notice templates and Escalate, deadline text, the PDF packet
Both	Room schema, Compose navigation, integration, testing

How we work: one GitHub repository with `main` protected, feature branches, and every pull request reviewed by the other person. Tasks are tracked in GitHub Issues and Projects. We talk on WhatsApp daily, have two short calls a week, and switch scrum master every week.

Update 1 on November 10 is a working slice: the simulator feeds Room, the chart shows indoor readings against 68/64°F with outdoor weather, and a heat issue opens automatically.



milestones · Oct 20 to Dec 8, 2026

Update 2 on November 24 is a feature-complete beta; the final app, paper and an 8 to 10 minute demo are due December 8.

Risks

The biggest risk is Android stopping Bluetooth in the background, and the simulator keeps the project moving if that happens.

Risk	Type	How we handle it
Android stops BLE sampling in the background	Technical	Foreground service, gaps flagged in the data, simulator as the demo path
The sensor arrives late or its data format is hard to read	Schedule	Simulated-sensor mode from week one
Legal text read as legal advice	Product	Cite mass.gov and MassLegalHelp; "not legal advice" on every letter and PDF
Chart work slipping	Schedule	Vico with sample data in week one; the PDF reuses the same data
Open-Meteo down during a demo	Technical	Last weather cached in Room; indoor chart still works
Lost or broken phone loses photos	Product	Export of the evidence folder; encrypted backup is a stretch goal
Too much scope for two people	Schedule	Strict tiers; nothing after the MVP starts until it works
Privacy of addresses and photos	Privacy	Photos stay on the phone, hashes only on the server, no analytics SDKs

References

1. 105 CMR 410.180, Temperature Requirements, Massachusetts State Sanitary Code
2. MGL c.111 §127L, repairs by tenants after certified violations
3. MassLegalHelp, Legal Tactics: Tenants' Rights in Massachusetts, Ch. 8 "Getting Repairs Made"
4. Mass.gov, Massachusetts law about winter heating
5. Open-Meteo weather API

6. Android Developers documentation: Jetpack Compose, Navigation, Room, WorkManager, CameraX, Bluetooth LE, foreground services